

1 Details of the baseline’s hyperparameters

For reproducibility, we introduce the best hyperparameter configurations for each dataset. We conducted experiments under the following hyperparameters:

- **FEARec**: Sampling Ratio $\alpha \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$, Auto-correlation Top- $k \in \{1, 3, 5\}$.
- **BSARec**: $\alpha \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$, $c \in \{1, 3, 5, 7, 9\}$.
- **MUFFIN**: the number of frequency bands $k \in \{2, 4, 6, 8, 10\}$, the auxiliary loss weight $\alpha \in \{0.05, 0.1, 0.2, 0.5, 1.0\}$, the balancing loss weight $\beta \in \{0.05, 0.1, 0.2, 0.5, 1.0\}$.
- **TiSASRec**: the attention heads are selected from $\{2, 4, 8\}$, time intervals are selected from $\{100, 256, 300, 400, 500\}$.
- **Mamba-based**: including Mamba4Rec, SIGMA, and TIM4Rec, we set the following parameters according to the original paper: the SSM state expansion factor is set to 32, the local convolution width is set to 4, and the block expansion factor is set to 2.
- **Transformer-based**: including SASRec, TCPSRec, and LinRec, we set the Transformer layer to 2, and attention heads are selected from $\{2, 4, 8\}$.

The details of the optimal parameters for the baseline on each dataset are as follows:

Table 1. Best hyperparameters of FMLPRec on all datasets

	Beauty	Video	Electronics	LastFM
Number of layers ℓ	1	2	1	2

Table 2. Best hyperparameters of FEARec on all datasets

	Beauty	Video	Electronics	LastFM
Top- k	5	3	3	3
Sampling Ratio α	0.7	0.5	0.7	0.7

Table 3. Best hyperparameters of BSARec on all datasets

	Beauty	Video	Electronics	LastFM
α	0.7	0.7	0.3	0.9
c	5	5	3	1
attention heads h	1	2	2	1

Table 4. Best hyperparameters of MUFFIN on all datasets

	Beauty	Video	Electronics	LastFM
k	6	6	4	6
α	0.1	0.1	0.1	0.1
β	0.2	0.2	0.2	0.2

Table 5. Best hyperparameters of TiSASRec on all datasets

	Beauty	Video	Electronics	LastFM
attention heads	4	4	4	8
time intervals	256	256	256	256

Table 6. Best hyperparameters of Mamba-based on all datasets. The three models have identical parameter configurations across the four datasets, so we have omitted the data rows.

	SSM state expansion factor	local convolution width	block expansion factor
Mamba4Rec	32	4	2
SIGMA and	32	4	2
TIM4Rec	32	4	2

Table 7. Best hyperparameters of Transformer-based on all datasets

	attention heads			
	Beauty	Video	Electronics	LastFM
SASRec	4	4	4	4
TCPSRec	4	4	4	4
LinRec	4	4	4	8